HUMAN ANATOMY STUDY GUIDE – UNIT 1 LECTURE EXAM

Directions: Use your **notes**, **PowerPoints**, and **book** to answer the following questions. Do NOT use the Internet – it will not prepare you as well for the exam.

Note: You may be asked for the same answer in different ways throughout the guide, which means you may have the ***same answer for multiple different questions****. This is not meant to confuse you, but rather to emphasize importance.*

FOUNDATIONS OF ANATOMY (Lesson 01)

1. Define the following terms:
   - Anatomy
   - Congenital
   - Homeostasis
   - Physiology
   - Prone
   - Supine

2. Be able to correctly use the following *directional* terms:
   - Superior
   - Inferior
   - Anterior
   - Posterior
   - Medial
   - Lateral
   - Proximal
   - Distal
   - Superficial
   - Deep

3. Be able to describe the following *planes,* and know which one is evaluated during a CT scan:
   - Frontal
   - Sagittal
   - Transverse

4. Define the following *regional* terms:
   - Abdominal
   - Antebrachial
   - Antecubital
   - Axillary
   - Brachial
   - Buccal
   - Carpal
   - Cephalic
   - Cervical
   - Coxal
   - Crural
   - Femoral
   - Frontal
   - Inguinal
   - Manus
   - Mental
   - Nasal
   - Oral
   - Palmar
   - Patellar
   - Pedal
   - Pelvic
   - Plantar
   - Pubic
   - Sternal
   - Tarsal
   - Thoracic
   - Umbilical

5. Know the levels of organization within the body, *specifically from smallest to largest*.

6. How do we obtain electrolytes?

7. What is the smallest living unit of the body?

8. Know what structures make up the **axial body** and **appendicular body**.
   - Which one involves the body trunk?
   - Which one involves the arms and legs?

9. Know what structures make up the **somatic body** and **visceral body**.
    - Which one involves superficial structures, like the skin and skeletal muscle?
    - Which one involves internal organs?

10. Know the 4 abdominal quadrants, and name an organ within each quadrant.

11. Understand the *basic* functions of the 11 organ systems (hint: *look at the red text on each organ system slide*).

BODY CHEMISTRY (Lesson 02)

12. Define the following terms:
    - Homeostasis
    - Hydrophilic
    - Hydrophobic
    - Hypertonic
    - Hypoglycemia
    - Hypotonic
    - Isotonic
    - Osmosis
    - Solute
    - Solvent

13. Why is water so good at dissolving things?

14. Know the basic features of carbohydrates, lipids (fats), proteins, and nucleic acids.
    - Which group is involved with energy and metabolism?
    - Which group is united by its **hydrophobia**?
    - Which group is responsible for cellular function?
    - Which group provides instructions on how to make proteins?
    - What groups (and subgroups) are found in a regular human diet?

15. Know the 4 types of **carbohydrates**.
    - Which one is associated with ATP production?
    - Which one is associated with *storage* in the **liver** and **skeletal muscle**?
        - How is the limited storage capacity in newborns significant?
    - Which one is associated with energy storage in plants?
    - Which one is associated with cell wall structure?

16. Know the 3 types of **lipids**.
    - Which type is found in the diet?
    - Which type is found in cell membranes?
    - Which type is found in cell membranes and hormones?
    - Which type has a hydrophilic portion and a hydrophobic portion?
    - Which type has a 4-ring structure?
    - Which type is used for energy?

17. Be able to give examples of proteins in the body.

18. Understand the basics of electrolytes:
    - Where can you find electrolytes with respect to the cell?
    - What do electrolytes do?
    - What do electrolytes control?

19. Understand how molecules like to *diffuse*.
    - Do they move from high to low concentrations or low to high?

20. Understand the basic principles of **osmosis**.
    - How does water move with respect to solute concentrations?
    - What would happen to a cell placed in a very *concentrated* environment?
    - What would happen to a cell placed in a very *dilute* environment?
    - Why can't humans drink salt water?

- What does water intoxication do to human cells?

21. What are the 2 major functions of the plasma membrane?

22. What molecule makes up the majority of the plasma membrane?

23. What types of molecules can cross the plasma membrane easily and *without* help?

24. What does **Active Transport** do, and what does it require?
    - Be able to recognize what scenarios require active transport

25. Know the 3 types of **bulk transport** (endocytosis).
    - Which type is known as “cellular-eating”?
    - Which type is known as “cellular-drinking”?
    - Which type is very specific for the molecules it chooses to let into the cell?

<u>THE CELL AND CELL DIVISION (Lesson 03)</u>

26. Name the 3 types of intercellular junctions:
    - Which one completely fuses cells together without creating a connection?
    - Which one does not completely fuse cells together?
    - Which one provides a passageway between cells?
    - Which one creates the strongest connection?
    - Which one prevents substances from passing between cells?
    - Which one allows cells to communicate with one another?

27. Understand the types of changes a cell can undergo:
    - Which change results in an *increased* <u>number</u> of cells in an area?
    - Which change results in an *increased* <u>size</u> of cells in an area?
    - Which change results in *decreased* cell <u>size</u> or <u>number</u> in an area?

28. Know the 2 mechanisms of cell death:
    - Which one is programmed cell death?
    - Which one occurs after severe damage?
    - Which one causes cellular contents to spill out of the cell?
    - Which one prevents cellular contents from spilling out of the cell?
    - Which one triggers the immune system and the inflammatory response?

29. Know the 3 basic components of the cell.

30. What cellular structure is considered "intracellular fluid"?

31. Know the basic functions of the following organelles (<u>hint</u>: *look at the red text on each organelle slide*):
    - Centrosomes
    - Cytoskeleton
    - Golgi Apparatus
    - Lysosome
    - Mitochondria
    - Nucleus
    - Nucleolus
    - Peroxisome
    - Plasma Membrane
    - Ribosome
    - Rough Endoplasmic Reticulum
    - Smooth Endoplasmic Reticulum

32. What is the advantage in mitochondrial membrane foldings?

33. What is different about the proteins produced by the cytoplasmic ribosomes vs. rough endoplasmic reticulum ribosomes?

34. Know the basic function of enzymes.

35. Know the 3 types of fibers that make up the cytoskeleton:
    - Which fiber is the smallest?
    - Which fiber is the largest?
    - Which fiber is found in the mitotic spindle?
    - Which fiber is found in flagella?
    - Which fiber is found in cilia?

- Which fiber is found in microvilli?

36. Know the 4 variations of the cytoskeleton:
    - Which variation produces beating movement?
    - Which variation produces directional movement?
    - Which variation increases surface area?
    - Which variation participates in cell division?

37. DNA is organized into units called ______________________.

38. What are histones, and why are they advantageous?

39. What is produced during <u>transcription</u>? What is produced during <u>translation</u>?

40. Know the flow of information regarding protein synthesis.
    - *In other words, _______ → _______ → ________ → Protein*
    - Which of the structures you identified above is ultimately responsible for a genetic disease, like Sickle Cell Anemia?

41. Know the 2 phases of the Cell Cycle
    - In which one does the cell spend most of its time?
    - Which one involves cell division?

42. Compare and contrast homologous chromosomes and sister chromosomes.
    - Which ones are identical?
    - Which ones are non-identical
    - Which ones did you inherit from your parents?
    - Which ones are always present in every cell?
    - Which ones only exist during cell division?
    - Which ones are pulled apart during cell division?

43. Name the 3 stages of Interphase

44. Know the 4 stages of Mitosis and what happens in each.
    - During what phase is the mitotic spindle formed?
    - During what phase do chromosomes line up along the equator of the cell?
    - During what phase does the nuclear membrane break down?
    - During what phase are sister chromosomes actually pulled apart?
    - During what phase do 2 separate nuclei begin to form?

45. During what stage of the cell cycle is DNA copied and sister chromosomes formed?

<u>EMBRYOLOGY and TISSUES PART I (Lesson 04)</u>

46. Define the following terms:
    - Differentiation
    - Gastrulation
    - IPSC
    - Neurulation
    - Pluripotent
    - Stem cell
    - Teratogen
    - Totipotent
    - Zygote

47. What are the 3 stages of the prenatal period?
    - During what stage does organogenesis occur?
    - Which stage is the longest?
    - Which stage is the shortest?

48. What are the 3 embryonic germ layers?
    - What is the name of the stage when they form?
    - Which one is the outer layer?
    - Which one is the middle layer
    - Which one is the inner layer?
    - Which layer forms the nervous system?
    - Which layer forms the digestive system?
    - Which layer forms the respiratory system?
    - Which layer forms the somatic body?
    - Which layer forms the cardiovascular system?
    - Which layer forms the skin?

49. What genes regulate the cell cycle?
    - What genes *stimulate* the cell cycle by encouraging cell division?
    - What genes *inhibit* the cell cycle by discouraging cell division?

50. Name 3 proteins made by cell cycle genes:
    - What protein stimulates cell division?
    - What proteins inhibit cell division?
    - What protein is made by proto-oncogenes?
    - What proteins are made by tumor-suppressor genes?
    - What proteins are on plasma membranes that prevent over-crowding?
    - What proteins connect cells to the body?

51. What disease can occur when genes controlling the cell cycle are abnormal?

52. What are mutated versions of genes that stimulate the cell cycle *too much* and cause cancer?

53. What are mutated versions of genes that don’t work to inhibit the cell cycle and cause cancer as a result?

54. What are the 2 functions of **epithelial tissue**?

55. Know the 4 features common to all epithelial cells.

56. Why is epithelial tissue so effective at healing itself?

57. Understand the basic features of **simple** epithelium.
- How many layers of cells are there?
- Does this make a good barrier?
- Is this tissue good at participating in organ activity?
- In what parts of the body might you find simple epithelium?

58. Understand the basic features of **stratified** epithelium.
- How many layers of cells are there?
- Does this make a good barrier?
- Is this tissue good at participating in organ activity?
- In what parts of the body might you find stratified epithelium?

59. Describe the <u>shape</u> of the following *epithelial* cell types:
- Squamous
- Cuboidal
- Columnar
- Pseudostratified columnar
- Transitional

60. What tissue type has cells with cilia, microvilli, or keratin?
- What cell shape can have cilia?
- What cell shape can have microvilli?
- Which cell shape and layering can have keratin?

TISSUES PART II (Lesson 05)

61. Name the 4 tissue types:
    - Which type supports the body?
    - Which type participates in an organ's function?
    - Which type conducts electrical signals?
    - Which type creates a barrier?
    - Which type produces movement?

62. Know the 3 types of membranes.
    - Membranes are usually which of the 4 tissue types?
    - What type of membrane lines cavities *open* to the exterior?
    - What type of membrane lines cavities *closed* to the exterior?

63. What is the main function of **connective tissue**?

64. What are the 3 components of all connective tissues?

65. Describe the 2 types of cell populations found in connective tissues.

66. Name the 4 types of protein fibers that can be found in connective tissues.

67. Describe the 5 classifications of connective tissues, and any relevant subcategories.

68. What is the main function of **muscle tissue**?

69. Describe the 3 types of muscle tissue:
    - What type is under conscious control?
    - What type is not under conscious control?
    - What types have striations in the cytoplasm?
    - What type does not have striations in the cytoplasm?
    - What type of muscle is found in the heart?
    - What type is found in blood vessels?
    - What type is found in the digestive tract?
    - What type is attached to the skeleton?

70. What is the main function of **nervous tissue**?

71. Name the 2 types of cells in nervous tissue.

72. Which of the 4 tissue types:
    - Is involved with *secretion*, *absorption*, *filtration*, and any other action an organ may perform?
    - Is the least cellular?
    - Has striations in its cytoplasm?
    - Is found in tendons and ligaments?
    - Is blood?